

Dashboard Handoff Instructions

Live dashboard package for the Verilog simulation of the Smart Agriculture Edge Analytics IP. The dashboard can be built and tested before RTL integration; the only fixed contract is the CSV row order.

Files To Send

- edge_agri_dashboard.py - the actual dashboard.
- demo_stream.py - fake data generator for testing before RTL is ready.
- VERILOG_DASHBOARD_CONTRACT.md - exact data format and Verilog print snippet.
- ROBOCHIPX_Dashboard_Instructions.pdf - this handoff guide.

1. Install

- Python 3.10 or newer from python.org.
- Icarus Verilog, so iverilog and vvp work in the terminal.
- VS Code extensions: Python and a Verilog/SystemVerilog extension.

No pip install is needed. The dashboard uses only Python built-in modules, especially tkinter.

```
python -m tkinter
```

2. Test Dashboard Without Verilog

```
python demo_stream.py | python edge_agri_dashboard.py
```

This opens the live dashboard with changing graphs, sensor timestamp, dashboard receive time, actuator lamps, alerts, crop health, and SAFE/WARNING/CRITICAL status.

3. Add This To The Verilog Testbench

Print the CSV header once:

```
initial begin
    $write("timestamp,moisture_raw,nutrient_raw,temp_raw,");
    $write("moisture_avg,nutrient_avg,temp_avg,pump_on,");
    $write("dose_nutrient,alert_nutrient,alert_weed,");
    $write("alert_heat,alert_frost,alert_anomaly,");
    $display("status,crop_health,relocate_recommend");
end
```

Print one row whenever the chip pipeline has a valid processed sample:

```
always @(posedge clk) begin
    if (sample_valid) begin
        $display("%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d,%0d",
            timestamp,
            moisture_raw, nutrient_raw, temp_raw,
            moisture_avg, nutrient_avg, temp_avg,
            pump_on, dose_nutrient, alert_nutrient, alert_weed,
            alert_heat, alert_frost, alert_anomaly,
            status,
            crop_health,
            relocate_recommend
        );
        $fflush;
    end
end
```

Use numeric status: 0 = SAFE, 1 = WARNING, 2 = CRITICAL.

4. Run With Real Simulation

Compile in VS Code terminal or PowerShell:

```
iverilog -o simulation.vvp `
    edge_analytics_top_tb.v edge_analytics_top.v `
    sensor_collector.v moving_avg.v `
    analytics_engine.v output_analytics.v
```

Run the simulation into the dashboard:

```
vvp simulation.vvp | python edge_agri_dashboard.py
```

5. Frozen CSV Contract

Field	Meaning
timestamp	Sensor/sample timestamp, sample counter, or \$time.
moisture_raw, nutrient_raw, temp_raw	Raw sensor values.
moisture_avg, nutrient_avg, temp_avg	Moving-average filtered values.
pump_on	Irrigation actuator output.
dose_nutrient	Fertilizer actuator output.
alert_nutrient, alert_weed	Low nutrient and resource-theft/weed alerts.
alert_heat, alert_frost, alert_anomaly	Climate and sensor fault alerts.
status	0 SAFE, 1 WARNING, 2 CRITICAL.
crop_health	0-100 fused plant-health score.
relocate_recommend	1 when the environment remains critical despite action.

Important

You do not need the Verilog code first. The dashboard can be built first using demo_stream.py. Later, the testbench only needs to print the same fields in the same order.